Circles

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Inequalities in one triangle**

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 14 cm. Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

2 **[Foundational]** • **Target: Inequalities in one triangle**

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 10. Substituting $x = 1$ into $y = 4x + 6$ gives $y = 4(1) + 6 = 10$.

3 **[Foundational]** • **Target: Inequalities in one triangle**

In ΔABC studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 85°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 52^\circ) = 85^\circ$.

4 **[Foundational]** • **Target: Inequalities in one triangle**

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 21 cm. Multiplying the original length by the scale factor gives $7 \times 3 = 21$ cm.

5 **[Foundational]** • **Target: Inequalities in one triangle**

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 14. Substituting $x = 2$ into $y = 4x + 6$ gives $y = 4(2) + 6 = 14$.

6 **[Foundational]** • **Target: Inequalities in one triangle**

In ΔABC studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 78°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 59^\circ) = 78^\circ$.

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7 [Foundational] • Target: Inequalities in one triangle

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 28 cm. Multiplying the original length by the scale factor gives $7 \times 4 = 28$ cm.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Inequalities in one triangle

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 18. Substituting $x = 3$ into $y = 4x + 6$ gives $y = 4(3) + 6 = 18$.

9 [Intermediate] • Target: Inequalities in one triangle

In ΔABC studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 71°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 66^\circ) = 71^\circ$.

10 [Intermediate] • Target: Inequalities in one triangle

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 35 cm. Multiplying the original length by the scale factor gives $7 \times 5 = 35$ cm.

11 [Intermediate] • Target: Inequalities in one triangle

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Final Answer: 22. Substituting $x = 4$ into $y = 4x + 6$ gives $y = 4(4) + 6 = 22$.

12 [Intermediate] • Target: The triangle inequality

In ΔABC studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 91°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (44^\circ + 45^\circ) = 91^\circ$.

13 [Intermediate] • Target: The triangle inequality

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 16 cm. Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

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14 **[Intermediate]** • **Target: The triangle inequality**

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 12. Substituting $x = 1$ into $y = 5x + 7$ gives $y = 5(1) + 7 = 12$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: The triangle inequality** • **Cross-Lesson: 11.01 Probability**

In ΔABC studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 84°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (44^\circ + 52^\circ) = 84^\circ$.

16 **[Challenging]** • **Target: The triangle inequality** • **Cross-Lesson: 2.01 Lines & Angles**

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 24 cm. Multiplying the original length by the scale factor gives $8 \times 3 = 24$ cm.

17 **[Challenging]** • **Target: The triangle inequality** • **Cross-Lesson: 3.01 Triangles & Congruence**

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 17. Substituting $x = 2$ into $y = 5x + 7$ gives $y = 5(2) + 7 = 17$.

18 **[Challenging]** • **Target: The triangle inequality** • **Cross-Lesson: 4.01 Similarity & Proportional Reasoning**

In ΔABC studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 77°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (44^\circ + 59^\circ) = 77^\circ$.

19 **[Challenging]** • **Target: The triangle inequality** • **Cross-Lesson: 5.01 Right Triangles & Trigonometry**

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 32 cm. Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

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20 [Challenging] • Target: The triangle inequality • Cross-Lesson: 6.01 Polygons & Quadrilaterals

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 22. Substituting $x = 3$ into $y = 5x + 7$ gives $y = 5(3) + 7 = 22$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Angle bisectors and perpendicular bisectors

In ΔABC studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 94° ✓

b. 104°

c. 86°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 45^\circ) = 94^\circ$.

22 [Foundational] • Target: Angle bisectors and perpendicular bisectors

A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm ✓

c. 20 cm

d. 45 cm

Multiplying the original length by the scale factor gives $5 \times 3 = 15$ cm.

23 [Foundational] • Target: Angle bisectors and perpendicular bisectors

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 8

b. 12

c. 10 ✓

d. 12

Substituting $x = 3$ into $y = 2x + 4$ gives $y = 2(3) + 4 = 10$.

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24 [Foundational] • Target: Angle bisectors and perpendicular bisectors

In ΔABC studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 73° ✓

b. 83°

c. 65°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 66^\circ) = 73^\circ$.

25 [Foundational] • Target: Altitudes, medians, and bisectors of triangles

In ΔABC studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 93° ✓

b. 103°

c. 85°

d. 138°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (42^\circ + 45^\circ) = 93^\circ$.

26 [Foundational] • Target: Altitudes, medians, and bisectors of triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

27 [Foundational] • Target: Altitudes, medians, and bisectors of triangles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$?

a. 11

b. 16

c. 14 ✓

d. 17

Substituting $x = 3$ into $y = 3x + 5$ gives $y = 3(3) + 5 = 14$.

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28 [Foundational] • Target: Altitudes, medians, and bisectors of triangles

In ΔABC studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 72° ✓

b. 82°

c. 64°

d. 138°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (42^\circ + 66^\circ) = 72^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Angle bisectors and perpendicular bisectors

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm ✓

c. 15 cm

d. 20 cm

Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

30 [Intermediate] • Target: Angle bisectors and perpendicular bisectors

In ΔABC studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 87° ✓

b. 97°

c. 79°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 52^\circ) = 87^\circ$.

31 [Intermediate] • Target: Angle bisectors and perpendicular bisectors

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm ✓

c. 25 cm

d. 80 cm

Multiplying the original length by the scale factor gives $5 \times 4 = 20$ cm.

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32 [Intermediate] • Target: Altitudes, medians, and bisectors of triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm ✓

c. 18 cm

d. 24 cm

Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

33 [Intermediate] • Target: Altitudes, medians, and bisectors of triangles

In ΔABC studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 86° ✓

b. 96°

c. 78°

d. 138°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (42^\circ + 52^\circ) = 86^\circ$.

34 [Intermediate] • Target: Altitudes, medians, and bisectors of triangles

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm ✓

c. 30 cm

d. 96 cm

Multiplying the original length by the scale factor gives $6 \times 4 = 24$ cm.

35 [Intermediate] • Target: Inequalities in one triangle

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm ✓

c. 21 cm

d. 28 cm

Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

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36 [Intermediate] • Target: Inequalities in one triangle

In ΔABC studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 85° ✓

b. 95°

c. 77°

d. 137°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 52^\circ) = 85^\circ$.

37 [Intermediate] • Target: Inequalities in one triangle

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm ✓

c. 35 cm

d. 112 cm

Multiplying the original length by the scale factor gives $7 \times 4 = 28$ cm.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The triangle inequality • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm ✓

c. 24 cm

d. 32 cm

Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

39 [Challenging] • Target: The triangle inequality • Cross-Lesson: 5.01 Right Triangles & Trigonometry

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 12

b. 19

c. 17 ✓

d. 22

Substituting $x = 2$ into $y = 5x + 7$ gives $y = 5(2) + 7 = 17$.

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40 [Challenging] • Target: The triangle inequality • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm ✓

c. 40 cm

d. 128 cm

Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

41 [Challenging] • Target: Extension: Geometric mean • Cross-Lesson: 8.01 Area, Surface Area & Volume

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm ✓

c. 12 cm

d. 16 cm

Multiplying the original length by the scale factor gives $4 \times 2 = 8$ cm.

42 [Challenging] • Target: Extension: Geometric mean • Cross-Lesson: 9.01 Transformations & Symmetry

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 10

b. 14

c. 12 ✓

d. 14

Substituting $x = 2$ into $y = 2x + 8$ gives $y = 2(2) + 8 = 12$.

43 [Challenging] • Target: Extension: Geometric mean • Cross-Lesson: 1.01 Foundations & Reasoning

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm ✓

c. 20 cm

d. 64 cm

Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

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44 [Challenging] • Target: Dividing a segment into a given ratio • Cross-Lesson: 10.01 Coordinate Geometry

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ **and** $x = 4$ ✓

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

45 [Challenging] • Target: Dividing a segment into a given ratio • Cross-Lesson: 11.01 Probability

For the model $f(x) = 3x + 9$ in dividing a segment into a given ratio, evaluate $f(4)$.

a. 21 ✓

b. 24

c. 12

d. 25

Substitute $x = 4$: $f(4) = 3(4) + 9 = 21$.

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